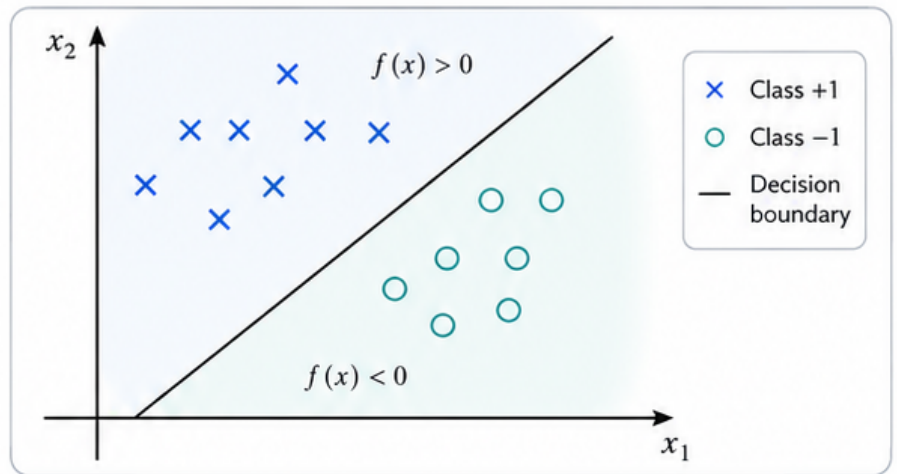


Linear Classifiers

Clean notes from handwritten lecture pages

1. Overview

- A linear classifier separates classes using a straight decision boundary (a hyperplane).
- Logistic regression is a common linear classifier.



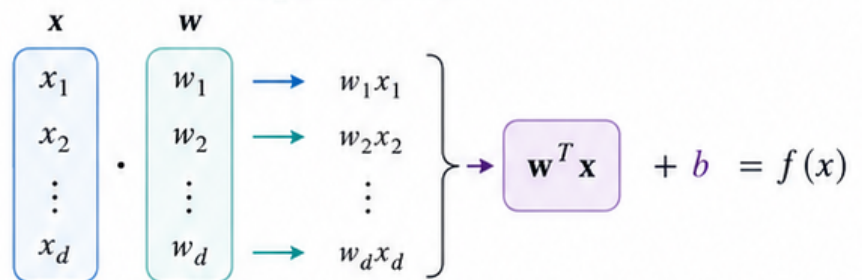
2. Model (Decision Function)

$$f(x) = \mathbf{w}^T \mathbf{x} + b$$

For two features (x_1, x_2) :

$$f(x) = w_1 x_1 + w_2 x_2 + b$$

Weighted sum view



The model computes a weighted sum of features, plus a bias.

3. Prediction Rule

If $f(x) \geq 0$, predict **+1**

If $f(x) < 0$, predict **-1**



Interpretation

The sign of $f(x)$ determines which side of the decision boundary the point falls on.

4. Multiclass Classification

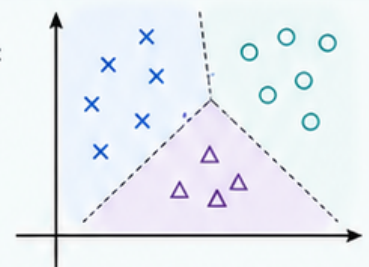
For K classes, the model produces one score per class:

$$s_k(x) = \mathbf{w}_k^T \mathbf{x} + b_k, \quad k = 1, \dots, K$$

Predict the class with the highest score:

$$\hat{y} = \underset{k}{\operatorname{argmax}} s_k(x)$$

This generalizes the idea: each class has a score function, and we pick the best one.



Key Takeaway: Linear classifiers use a weighted sum of features to produce a score. The sign (or highest score) determines the predicted class.

SGD Classifier and Hinge Loss

Learn a linear classifier by updating one sample at a time

i 1. Overview

SGDClassifier learns a linear classifier by updating the parameters one training example at a time.

Σ 2. Hinge Loss

$$L(w, b) = \max(0, 1 - y_i (w^T x_i + b))$$

The hinge loss is zero when the example is correctly classified with margin at least 1; otherwise it is positive.

⚖ 3. Margin Condition

Let $f(x_i) = w^T x_i + b$.

If $y_i f(x_i) < 1$, the example lies on the wrong side of the margin (or within the margin) → update the parameters.

Otherwise ($y_i f(x_i) \geq 1$), the example is correctly classified with sufficient margin → no update.

⚙ 4. Update Rules (Per Example)

$$f(x_i) = w^T x_i + b$$

$$w \leftarrow w + \eta y_i x_i$$

$$b \leftarrow b + \eta y_i$$

learning rate: $\eta = 0.1$



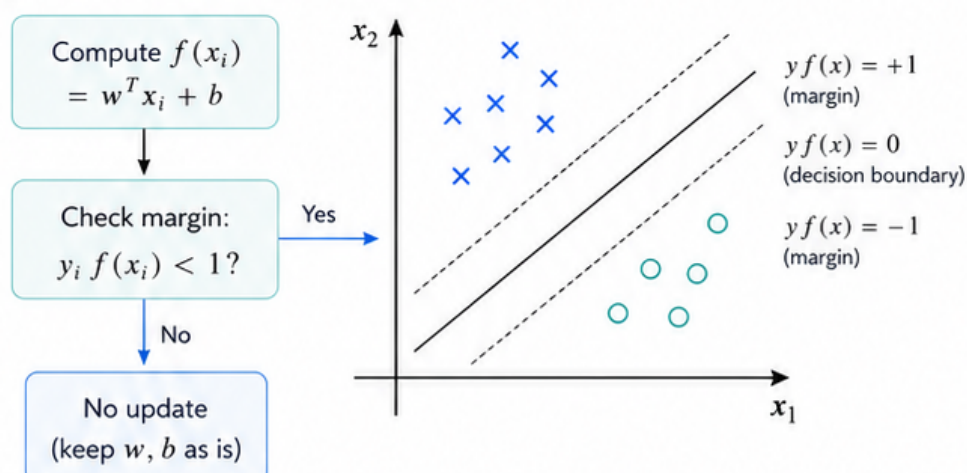
We adjust w in the direction of $y_i x_i$ and shift b in the direction of y_i .

📋 5. Training Data

x_1	x_2	y
2	1	+1
1	2	+1
-1	-1	-1
-2	-1	-1
1	-1	+1

Samples are processed sequentially in the order shown.

🔗 6. Margin Check → Update / No Update



Key Takeaway

SGDClassifier uses the hinge loss to find a linear decision boundary. - It checks the margin for each sample and updates the parameters only when the margin condition is violated.

Worked Example: Samples 1-3



Initial values: $w_1 = 0$, $w_2 = 0$, $b = 0$, $\eta = 0.1$



Reminder

$$f(x) = w_1 x_1 + w_2 x_2 + b$$

update when $y_i f(x_i) < 1$

1 Sample 1

$$x = (2, 1), y = +1$$

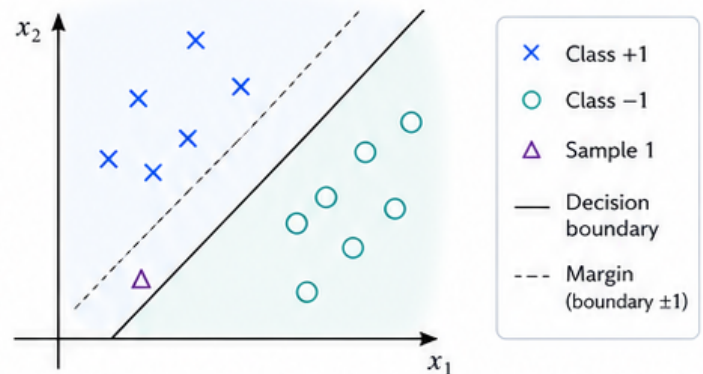
Compute: $f = (0)(2) + (0)(1) + 0 = 0$

Margin: $yf = (+1)(0) = 0 < 1 \rightarrow$ **update needed**

Update rule (for $y_i f(x_i) < 1$):

$$w \leftarrow w + \eta y x, \quad b \leftarrow b + \eta y$$

New parameters: $w_1 = 0.2$, $w_2 = 0.1$, $b = 0.1$



2 Sample 2

$$x = (1, 2), y = +1$$

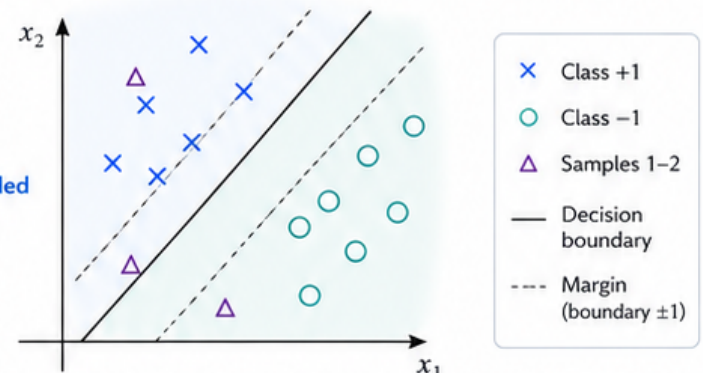
Compute: $f = (0.2)(1) + (0.1)(2) + 0.1 = 0.5$

Margin: $yf = (+1)(0.5) = 0.5 < 1 \rightarrow$ **update needed**

Update rule (for $y_i f(x_i) < 1$):

$$w \leftarrow w + \eta y x, \quad b \leftarrow b + \eta y$$

New parameters: $w_1 = 0.3$, $w_2 = 0.3$, $b = 0.2$



3 Sample 3

$$x = (-1, -1), y = -1$$

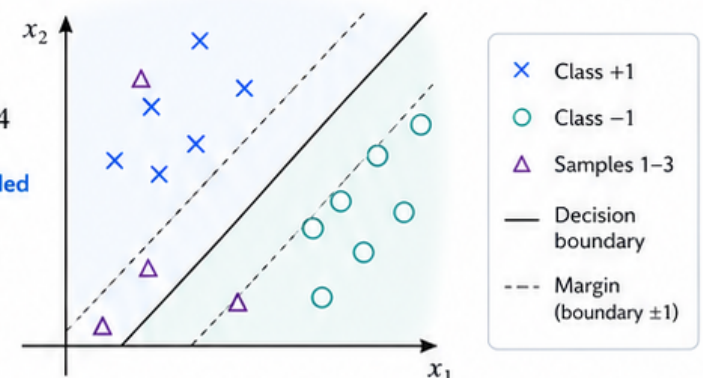
Compute: $f = (0.3)(-1) + (0.3)(-1) + 0.2 = -0.4$

Margin: $yf = (-1)(-0.4) = 0.4 < 1 \rightarrow$ **update needed**

Update rule (for $y_i f(x_i) < 1$):

$$w \leftarrow w + \eta y x, \quad b \leftarrow b + \eta y$$

New parameters: $w_1 = 0.4$, $w_2 = 0.4$, $b = 0.1$



Checkpoint: After Sample 3:

$$w = (0.4, 0.4), \quad b = 0.1$$

These parameters define the current decision boundary after three SGD updates.

Samples 4-5 and Final Classifier

Complete the worked example and summarize the final model.



Reminder: Current parameters before Sample 4:

$$w_1 = 0.4, \quad w_2 = 0.4, \quad b = 0.1$$

4 Sample 4 $x = (-2, -1), y = -1$

- Compute score:

$$f = (0.4)(-2) + (0.4)(-1) + 0.1 = -1.1$$

- Margin:

$$y f = (-1)(-1.1) = 1.1 > 1$$

$\Rightarrow$ no update needed



No update needed.

Parameters remain:

$$w_1 = 0.4, \quad w_2 = 0.4, \quad b = 0.1$$

5 Sample 5 $x = (1, -1), y = +1$

- Compute score:

$$f = (0.4)(1) + (0.4)(-1) + 0.1 = 0.1$$

- Margin:

$$y f = (+1)(0.1) = 0.1 < 1$$

$\Rightarrow$ update needed



Update performed.

Final parameters:

$$w_1 = 0.5, \quad w_2 = 0.3, \quad b = 0.2$$



Final classifier:

$$f(x) = 0.5x_1 + 0.3x_2 + 0.2$$

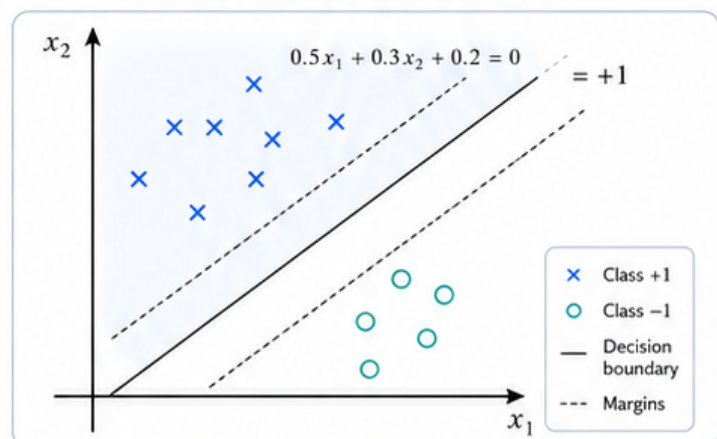
Prediction rule:

predict **+1** if $f(x) \geq 0$,
else predict **-1**.

Final classifier evaluation on training data

x_1	x_2	score $f(x)$	prediction
2	1	1.5	+1
1	2	1.3	+1
-1	-1	-0.6	-1
-2	-1	-1.1	-1
1	-1	0.4	+1

Decision boundary of final classifier



For a correctly classified point with margin > 1 , the hinge loss is 0.



Training accuracy:

100%